



Measurement of Temperature

The most important of the measurand variables encountered in industrial processes is Temperature. Because there are changes in the physical chemical state of most substances when they are heated or cooled. The measurement of temperature is very common in the field of instrumentation.

There are lot of devices available for the measurement of temperature such as

- Resistance thermometers
- Thermocouples
- Thermopiles
- Bimetallic thermometers
- Radiation pyrometers etc.
- Thermistors

The above mentioned devices can be classified into two groups.

- Contact Type
- Non-contact type (Radiation Type)

Contact Type:

In this category measuring device indicator is placed in direct contact with the heat source.

Ex:

- Resistance thermometers
- Thermistors
- Thermocouples
- Bimetallic Thermometers
- Thermopiles

Non-contact type (Radiation Type):

- In this category temperature measuring devices operate either on total radiation i.e. heat and light from the hot body or on the visible radiation.
- The devices operating on total radiation from the hot body are called the Radiation type

devices while those operating on visible radiation are called optical type devices.

- These temperature measuring devices are particularly suitable for the measurement of very high temperatures.

Resistance Thermometers (RTD)

- Resistance thermometer often referred to as resistance-temperature detector (RTD) operates upon the fact that almost all pure metals have the property of varying their resistance with temperature and change in resistance is almost directly proportional to the change in temperature.
- The range of temperature over which this is valid is decided by the temperature coefficient of resistance, chemical inertness and its crystal structure which should not undergo permanent changes with in this range.
- Electrical resistance with temperature, for most metallic materials can be represented by an expression.

$$R_t = R_0 [1 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \dots \alpha_n t^n]$$

Where

$R_0 \rightarrow$ Resistance in Ω at reference temperature (usually at ice point, 0°C)

$R_t \rightarrow$ Resistance in Ω at temperature t

$\alpha_1 \rightarrow$ Temperature coefficient of resistance in $\Omega/\Omega/^\circ\text{C}$

$\alpha_2, \alpha_3, \dots, \alpha_n \rightarrow$ Coefficients determined on the basis of two or more known resistance – temperature (calibration) points.

- The number of terms necessary depends on the material, the desired accuracy, and the range of operation.



- In the narrow range of operation α_2 and higher order coefficients are negligibly small.
 $\therefore$ The expression becomes
 $R_t = R_0[1 + \alpha_1 t]$
- The RTD is applicable for measurements of small temperature difference as well as for wide ranges of temperature.
- The RTD doesn't generate its own voltage, so a voltage source is required to be incorporated into the measuring circuit.
- The main draw-back of an RTD lies in its large size and sophisticated instrumentation.
- Wheat-stone Bridges are usually employed for measurement of variations in resistance, owing to changes in temperature, arising in RTDs.

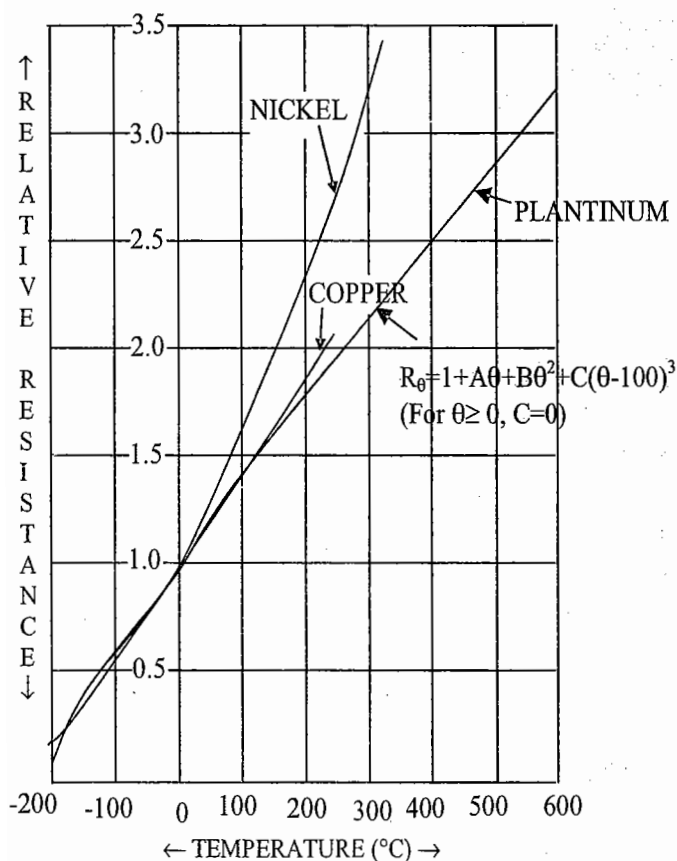
Requirements for the Resistance Materials used in RTDs:

- High temperature coefficient of resistance (in order to give substantial change in resistance for a relatively small change in temperature i.e. larger sensitivity.)
- High Resistivity (to ensure small wire length for high resistance or a larger resistance for the same wire length and so increased output).
- Linearity of relation between resistance and temperature (for convenience in measurement)
- Stability of the electrical characteristics of the material and resistance to contamination for good repeatability.
- Sufficient mechanical strength (so that resistance wire could be drawn very fine for reducing response time and at the same time
- adequate ruggedness in construction is obtained).
- Platinum, Nickel, and copper are the most commonly used resistance materials.
- Gold and silver are rarely used owing to their low resistivities.
- Tungsten has relatively high resistivity, but its use is limited for high temperature applications, as it is extremely brittle and difficult to work.
- Electrolytic copper of the highest purity is readily available in the market, having a highly consistent temperature coefficient of resistance which is slightly higher than that of platinum. But its use is limited to low-range industrial applications because of low resistivity and tendency toward oxidation at higher temperatures.
- In the range below 125°C, a gold-silver alloy having the same characteristics as platinum can be employed.
- For low temperature measurements certain Phosphor-Bronze alloys are found satisfactory.
- RTDs made out of Nickel wire can be employed with good stability and repeatability in temperature range of -70°C to 150°C .
- Nickel is less expansive than platinum and has a somewhat higher temperature coefficient and higher tensile strength but it is more non-linear than copper and above 300°C , it undergoes transformations which make its resistance-temperature curve irregular.
- Platinum is the most suitable material for RTDs most laboratory work and for industrial measurements of high accuracy over a wide temperature range.
- Platinum is commercially available in a pure form is relatively stable under different environmental conditions, and its resistance temperature curve is simple and holds over a wide range of temperatures (-263°C to $+545^\circ\text{C}$) with high precision.
- Its resistivity tends to increase less rapidly at higher temperatures than for the other metals (unique quality).
- It can withstand high temperatures while maintaining excellent stability.



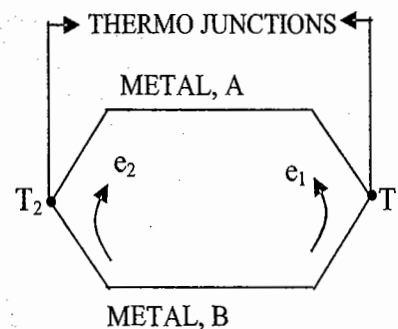
Materials used in RTDs:

Name of the material	Temperature coefficient of resistance/ °C	Temperature range		Accuracy	Melting point
		Min	Max		
Platinum	0.00391	-250°C	900°C	± 0.5°C	1775.5
Copper	0.00425	-200°C	150°C	± 0.2°C	1083
Nickel	0.0066	-70°C	150°C	± 0.2°C	1455
Tungsten	0.0045	-200°C	1000°C	-	3370

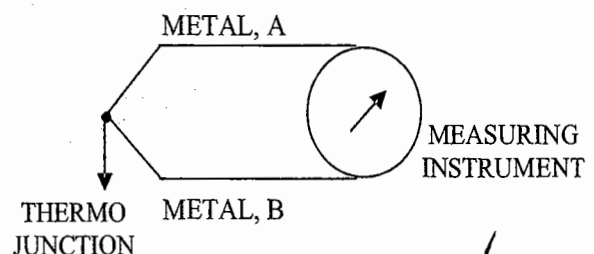


Thermo Couples

Thermocouples are perhaps the most simple and most widely used devices for measurement of temperature. It essentially consists of two dissimilar metal wires A and B, insulated from each other but welded or brazed together at their ends forming two junctions.



When one end of each wire is connected to a measuring instrument as shown in figure below it becomes an accurate and sensitive temperature measuring device.



The operation of a thermocouple is based on seeback effect.

Seeback effect:

If two wires of different metals are joined together at each end and form a complete electric circuit then current flows in the circuit when the two junctions are kept at different temperatures.



Thermo electric emf:

It is the emf which is responsible for current flow between two dissimilar metals in a thermocouple this emf is generated when junction temperatures difference should greater than zero.

The relation between the thermo-emf setup and temperature difference of hot and cold (or reference) junction is given by

$$E = \alpha (T - T_0) + \beta (T^2 - T_0^2)$$

Where,

E – thermo electric emf (volts)

T – absolute temperature of hot junction

T₀ – absolute temperature of cold junction

α , β - constants (depends upon the metals used)

∴ Thermo-electric emf is a function of temperature difference.

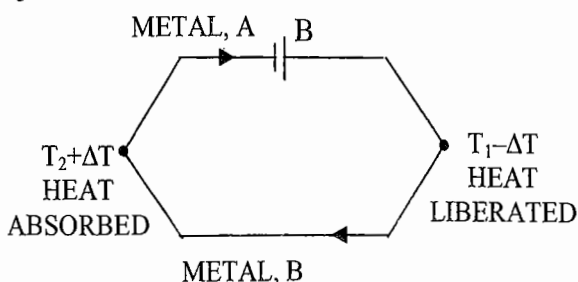
The emf developed by thermocouple depends only on material used and temperature of hot and cold junction. Not depends on shape, size length and area of material used and also method of joining them.

Reverse of seeback effect is also true it is explained Peltier effect.

Peltier effect:

Peltier observed that when an electric current flows across a junction of two dissimilar metals heat is either generated (liberated) or absorbed, depending upon the direction of flow of current.

When the electric current flows in the same direction as the seeback current, then that is absorbed at the hot Junction and liberated at the cold junction.



Another Reversible heat – flow effect is the Thomson effect.

Laws Governing thermocouples:

- Thermocouple circuits are normally required to be broken for introducing a galvanometer type instrument to measure the current or a known emf to balance the generated thermo-electric emf.
- In doing this additional metals and junctions are introduced, which may be at various temperatures therefore the following laws summarize the effects of these.

Law of Homogenous Metals:

A thermoelectric current cannot be sustained in a circuit of a single homogeneous material, however varying in cross section, by the application of heat alone. A consequence of this law is that two different materials are required for any thermocouple circuit.

Law of intermediate metals:

This law states that a third homogeneous material can always be added to a circuit with no effect on the net emf of the circuit as long as its extremities are at the same temperature.

This law makes it possible to use extension wires of a metal different from the thermocouple materials.

Law of intermediate temperatures:

If two dissimilar homogenous metals produce a thermo electric emf of E_1 when the junctions are at temperatures T_1 and T_2 , and a thermoelectric emf of E_2 when the junctions are at T_2 and T_3 , the emf generated with the junctions at temperatures T_1 and T_3 will be $E_1 + E_2$.

The application of this law permits a thermocouple calibrated for a given reference temperature to be employed with any other reference temperature through the use of a suitable correction.



Properties of thermocouples:

Type (ISA Standard)	Metal Alloys Used		Sensitivity in $\mu\text{V}/^\circ\text{C}$	Accuracy in $^\circ\text{C}$	Temperature Range in $^\circ\text{C}$	Spot Maximum in $^\circ\text{C}$
	Positive Wire	Negative Wire				
Type E	Chromel; 90% Nickel and 10% Chromium	Constantan; 57% Chromium and 43% Nickel	40 to 55	$\pm 1.8^\circ$ $\pm 0.5\%$	-200 to 850	1100
Type K	Chromel; 90% Nickel and 10% Chromium	Alumel 94% Ni, 3% Mn; 2% Al and 1% Si	40 to 55	$\pm 2.2^\circ$ $\pm 0.5\%$	-200 to +1100	1300
	Iron	Constantan; 57% Chromium and 43% Nickel	45 to 57	$\pm 2.2^\circ$ $\pm 0.5\%$	-200 to +850	1100
Type T	Copper	Constantan; 57% Chromium and 43% Nickel	15 to 60	$\pm 1.8^\circ$ $\pm 0.5\%$	-200 to +350	500
Type R	Platinum Rhodium; 87% Platinum and 13% Rhodium	Platinum	5 to 12	$\pm 0.25^\circ$	0 to 1000	1200
Type S	Platinum Rhodium 90% Platinum and 10% Rhodium	Platinum	5 to 12	$\pm 0.25^\circ$	450 to 1500	1650
	Rhodium – Iridium 60% Rhodium and 40% Iridium	Iridium	5	$\pm 1.0^\circ$	1200 to 2000	

Merits:

- Thermocouples are cheaper than resistance thermometers.
- The temperature range of thermocouples is about 1400°C . Cooled thermocouples can operate upto 4000°C .
- Except for the refractory group (tungsten-tungsten-rhenium) thermocouples are rugged in construction and can withstand high shock
- Thermocouples can be made in very small sizes.
- Thermocouples can be made from wire pairs as 0.013mm in diameter for millisecond

response. Special high speed thermocouples have time constant as small as $10\mu\text{s}$.

- Thermocouples can be made from heavy-gauge wire or even round stock to withstand the most severe applications.

Demerits:

- Thermocouples have less accuracy than RTDs (± 0.25 to $\pm 1\%$ reading depending on type and temperature range)
- Thermocouples need periodical checking when employed at elevated temperatures.



- Thermocouples need compensating leads as these are placed remote from measuring devices.
- Instrumentation must resolve terms of micro volts per degree of temperature.
- Reference junction compensation is required in thermocouples.

Applications:

- Thermocouple pyrometers, like resistance pyrometers can be applied to measurements at a number of points.
- Thermocouple, because of their small heat capacity and very small bulk are particularly suitable for indication of rapidly changing temperature and for use in localized and in otherwise accessible positions.
- Thermocouple particular application is in the measurement of surface temperature.

Thermistor

- Thermistors are also called the thermal resistors and the name is derived from thermally sensitive resistors, as the resistance of a thermistor varies as a function of temperature.
- Thermistors are essentially semiconductor devices, that behave as resistors with high negative temperature coefficient (usually - 0.04% at room temperature of 25°C) and are at least 10 times as sensitive as the platinum resistance thermometer.
- Thermistors are both positive temperature coefficient and negative temperature coefficient types, but most thermistors are negative temperatures coefficient types.
- Thermistors can detect very small changes in temperature which could not be observed with an RTD or a thermocouple.
- Sensitivity → Thermistor > RTD > Thermocouple

- Thermistors are widely used in applications which involve measurements in the range of - 60°C to 15°C.
- The resistance of thermistors ranges from 0.5Ω to 0.75Ω.
- Thermistors exhibits a highly non-linear characteristic of resistance versus temperature.
- The resistance R of a thermistor at a temperature T can be given by expression.

$$R = \alpha e^{\beta/T}$$

- Where α , β are constants depending upon the material and manufacturing techniques.
- The approximate relationship between resistance and temperature applying to most thermistors can be obtained by re-writing above equation (i) for temperatures T_1 and T_0 .

$$R_1 = R_0 e^{\beta(1/T_1 - 1/T_0)}$$

$\beta \rightarrow$ Thermistor constant, expressed in Kelvin and is of the order of 4000.

- $R_1, R_0 \rightarrow$ Resistance at absolute temperatures T_1 and T_0 .
- The assumption that β is constant gives a relation for the resistance temperature coefficient which is given as Temperature coefficient resistance

$$\alpha = \frac{dR_1}{dT} / R_1 = \frac{-\beta}{T^2}$$

- **Note:** From above equation we know that α is not constant. It actually drops with rise in temperature.

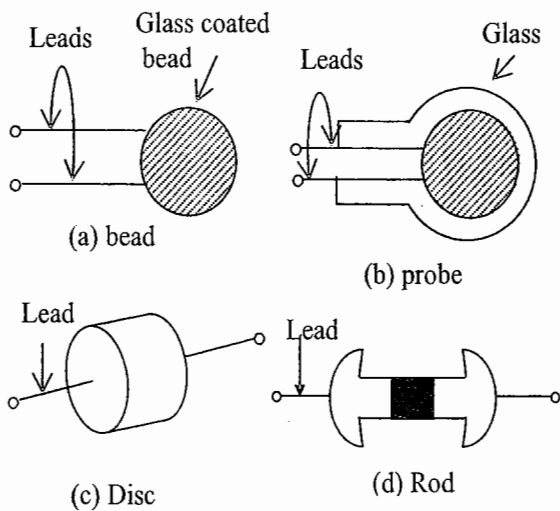
Ex: If $\beta = 4000$ and $T = 25^\circ\text{C}$ or 298 K then temperature coefficient of resistance

$$\alpha = \frac{-4000}{(298)^2} = -0.045 \Omega/\Omega/^\circ\text{C}.$$

Materials used for manufacturing thermistors:

Modern thermistors are manufactured from the oxides of metals like manganese, nickel, cobalt, copper, iron, zinc, aluminium titanium, magnesium and uranium.

Because of their high resistance they can be manufactured in very small sizes in the form of beads, discs and rods.



Note:

- If the thermistor is instead put into a wheat stone bridge the measuring system can detect temperature variations as small as $\pm 0.005^\circ\text{C}$.
- High resistance temperature coefficient characteristic of them results in inherent high sensitivity and high level of output, eliminating the need for extremely sensitive read out devices and lead wire matching techniques respectively.
- The high degree of non-linearity in the resistance-temperature function usually limits the range of the readout instrumentation.
- Calibration, stability and limitations an inter changeability are inherent to thermistors.
- Parallel and series resistors are employed for reducing non-linearity of thermistors and make thermistors easier to use over reasonably wide temperature range.
- The time constant varies directly with the thermal capacity of the thermistor and inversely with dissipation factor.

Features of thermistors:

- They are compact, rugged, and inexpensive.
- When properly aged they have good stability.
- The response time of thermistors can vary from a fraction of second to minutes depending on the size of the detecting mass and thermal capacity of the thermistor.
- The upper operating limit of temperature for thermistor is dependent on physical changes in the material or solder used in attaching the electrical connections and is usually 400°C or less.
- Thermistors can be installed at a distance from their associated measuring circuits if elements of high resistance are used such that the resistance of leads does not affect the readings and hence errors on this count are negligible.
- The average power dissipation can be effectively reduced and the highest sensitivity retained by energizing the thermistor with pulses of measuring power.

Applications:

Thermistors have a wide range of applications with the most common being direct temperature sensing and surge suppression. The characteristics of NTC and PTC thermistors lend themselves to applications including.

- Liquid level indicators
- Temperature compensation
- Flow measurement.
- Vacuum gages
- Thermal protection
- Amplifier-gain control
- Time delay circuits.
- Thermal switches etc....
- Measurement of gas composition.
- Measurement of difference of two temperatures.